

Post-Launch Monitoring & Observability Roadmap

A 6-week phased plan to keep the storefront and EPOS spotless after launch — covering uptime, security, SEO, link integrity, PDP health, and the **backbone admin features** (Bulk Category Editor, Supplier Products sync) that hold everything together.

Already live (day 1):	1. /api/health endpoint · 2. Sentry wiring (set DSN to activate) · 3. Security headers
Status:	All 3 verified passing — DB ping 2 ms, 6 OWASP headers applied site-wide.
Production sync:	Click "Save to Github" so Railway picks these up. Set SENTRY_DSN env var on Railway to activate.

Tier 1 — Always-on Watchdogs (smoke alarms)

These run 24/7 and only ping you when something breaks. The point isn't to stare at dashboards — it's to sleep at night.

What	Why	Tool / Approach	Status
Uptime monitor	Know within 60 s if site is down	UptimeRobot or BetterStack — pings /api/health every 1 min	TODO
Error tracking	Capture every JS + Python crash with stack trace	Sentry (free 5k events/mo)	■ Wired
API health endpoint	Single fast endpoint for monitors to ping	/api/health (DB ping + uptime)	■ Live
Real User Monitoring	Catch slow pages before reviews flag it	Cloudflare Web Analytics (free)	TODO
Daily UI Health PDF	18 selectors checked, emailed at 03:00 UTC	Already built	■ Live
Resend webhook handler	Track which transactional emails bounced	POST /api/email-webhook	TODO

Tier 2 — Security (lock the doors)

What	Why	How	Status
Security headers	CSP / HSTS / XFO / XCTO / Referrer-Policy / Permissions-Policy	FastAPI middleware	■ Live
Rate limiting	Stop brute-force on login + card-testing on checkout	slowapi: 5/min login, 10/min checkout	TODO
CSRF protection	Stop drive-by POSTs from malicious sites	Bearer tokens cover most; admin forms need tokens	TODO
Dependency CVE scan	Auto-flag vulnerable packages	GitHub Dependabot + weekly pip/yarn audit	TODO
Secret leak scan	Stop API keys being committed	gitleaks pre-commit hook	TODO
Admin audit log	Who logged in, when, from what IP	MongoDB collection + admin viewer	TODO

What	Why	How	Status
Webhook signature verify	Verify Stripe + Resend HMAC signatures	Already in Stripe SDK; audit Resend handler	Audit
2FA for admins	One stolen password ≠ entire site compromised	TOTP via pyotp	TODO

Note: Security headers were applied site-wide on day 1. The default CSP allows Stripe + GA + Resend + R2 image CDN. If a third-party widget breaks after deploy, set CSP_REPORT_ONLY=true in env to debug without blocking.

Tier 3 — Links & Navigation (nothing broken anywhere)

What	Why	How	Frequency
Internal link crawler	Walks every link from homepage + sitemap, depth-2 follow, flags 404 / 500 / redirect loops	Daily cron, headless Playwright (already installed)	Daily
Sitemap.xml validator	All URLs return 200, lastmod current, robots.txt allows them	Daily cron	Daily
External link checker	Manufacturer specs and supplier portals go down often	Weekly cron	Weekly
Redirect chain audit	Old URL → new URL must 301 cleanly without chains	Part of crawler	Daily
Zero-results search log	Customers searching “60x60 marble” but seeing empty page = lost sale	Log + weekly digest	Weekly

Tier 4 — PDP Integrity (every product page perfect)

Check	Why it matters	Frequency
At least 1 image	Empty image = lost sale	Daily
Price > 0	£0.00 = customer confusion	Daily
Stock qty OR always_in_stock=true	Out-of-stock not flagged = oversell	Daily
Finish + Size (tiles only, NOT accessories)	Filters break without these — e.g. /tiles?finish=matt	Daily
Description ≥ 50 chars	SEO and customer trust	Weekly
Product images load (no 404)	Already covered for supplier sync — extend to live PDPs	Daily
schema.org Product markup valid	Required for Google rich snippets	Weekly
Add-to-cart button works	Already in UI Health checks	■ Live
Trade pricing displays correctly	Already covered (the recurring trade-login bug)	■ Live

Tier 5 — Website Structure & SEO

What	Why	Tool
Heading hierarchy audit	h1→h3 jumps confuse screen readers + Google	Weekly Playwright crawl
Alt text on images	WCAG accessibility + SEO	Same crawl
Meta title + description per page	Empty meta = Google guesses badly	Same crawl
Canonical tags	Stops duplicate-content penalties	Same crawl
Open Graph + Twitter Card tags	Social shares look professional	Spot-check
Page speed (Core Web Vitals)	LCP < 2.5 s, CLS < 0.1, INP < 200 ms — Google ranks on these	PageSpeed Insights API daily
Mobile-friendly score	60 %+ of UK tile shoppers on mobile	Daily PageSpeed mobile

Tier 6 — Backbone Internal Features (the engine room)

These are the admin features the entire system depends on. They have many small components (sync, scopes, staging, name-transformation, image upload) and silently breaking any one of them propagates through the storefront and EPOS. **Watch them like a hawk.**

6A · Bulk Category Editor

Health Check	What goes wrong if missed	Cadence
Save persists every field (incl. nested specs, tags, images)	Edits silently dropped	Per-save smoke test
Reorder doesn't drop categories	Categories vanish from menu	After every save
Bulk price update math	10 % markup applied twice	Daily aggregate spot check
Bulk stock update	Negative stock values accepted	Validation test
Image bulk swap rolls back on failure	Half-broken state	Per-batch transaction
Save latency < 2 s for 100 rows	Admin UX degradation	P95 latency dashboard
Concurrent-edit conflict warning	One admin overwrites another silently	Lock + last-modified header

6B · Supplier Products Sync

Health Check	What goes wrong if missed	Cadence
Last-sync timestamp per supplier	Stale catalogue, customers see old prices	Hourly + alert if > 24 h
Pending staging count	Backlog grows unnoticed	Daily digest
Failed-sync counter	Single supplier silently broken for days	Slack alert on threshold
Image upload to R2 success rate	Products published without images	Daily aggregate
Duplicate detection (SPLENDOUR_SERIES_TO_UNIQUE_NAME)	Same product from 2 suppliers gets same name	On every transform
Naming transformation success rate	Raw supplier names leak to PDP	Alert if < 95 %
finish + size populated for tile suppliers	Filters break, customer searches return empty	Block publish if missing
Cost-price markup not double-applied	Prices doubled, refund cascade	Per-product validator
Manual edits preserved across sync (price + name locks)	Admin work overwritten on next sync	Pre-sync audit
Excluded categories respected	Deleted categories reappear	Pre-sync filter
Non-tile-product exclusion	Adhesives appear in /shop/tiles	Filter at sync time

Recommended dashboard: a single "Sync Health" admin page that shows per-supplier status, last-sync age, last-failure reason, pending staging, and a "Test sync" button. Cron probes each supplier daily and writes a row to db.sync_health_log for trend graphs.

Tier 7 — Data Hygiene & Backups

What	Why	How
Daily MongoDB backup	Disaster recovery	Railway has this — verify retention ≥ 30 days
Weekly restore test	Untested backups = no backups	Quarterly: spin up staging from last backup
R2 image versioning	Recover deleted images	Cloudflare R2 versioning — enable it
Order data integrity	Total = sum(line items) + shipping + VAT	Already covered for invoice deposits
Orphan record cleanup	Stale carts > 30 days, expired tokens	Weekly cron

Tier 8 — Logs & Observability

What	Why	Tool
Centralised logs	Searchable across backend + frontend + Stripe webhooks	Logtail or BetterStack (free tier)
Slow-query log	MongoDB queries > 100 ms surfaced	Mongo profiler
API response-time tracking	P95 latency per endpoint	Sentry Performance (already installed)
Frontend bundle size watch	Catch when someone imports moment.js (200kb)	bundlesize CI check

Tier 9 — Alerting Strategy

The point of all the above is **not to stare at dashboards** — it's to get notified only when something is wrong. Three tiers:

Tier	Channel	Triggers
Pager (wakes admin)	SMS + Slack DM	Site down > 2 min · Stripe webhooks failing · DB unreachable · 5xx error rate > 1 % · Daily backup failed
Email digest (next morning)	Resend → admin inbox	Daily UI Health PDF (live) · Daily reconciliation (live) · Weekly: top 0-result searches, broken external links, low-stock items · Monthly: dependency CVE summary, refund rate, conversion funnel
Quiet (just log)	DB / Sentry breadcrumbs only	Successful crons · Normal traffic · Cache hits

Suggested 6-week post-launch rollout

Week	Build	ROI
1	Sentry DSN set ✓ + UptimeRobot account ✓ + /api/health verified ✓ + security headers ✓	Catches 80 % of incidents in < 1 min
2	Pre-Launch Scan button (next) + broken-link crawler + sitemap.xml validator	Prevents customer-facing 404s
3	Rate limiting + admin 2FA + Dependabot + gitleaks pre-commit	Locks the doors
4	PDP integrity daily cron + zero-results search log + Stripe failed-payment alert	Protects revenue
5	Sync Health dashboard (Tier 6) + Core Web Vitals daily check + accessibility audit	Protects backbone + SEO
6	Centralised logs + admin audit log + backup restore test + monthly review cadence	Forensic readiness

Total effort: roughly **30–40 focused hours** spread across 6 weeks. After Week 1 you're already protected from the most common launch-day failure modes (silent crashes, downtime, XSS).

Where to put the credentials

Service	Env var	Where to set it
Sentry (backend)	SENTRY_DSN	Railway → Variables → Add
Sentry (frontend)	REACT_APP_SENTRY_DSN	Railway → Variables → Add
Sentry env tag	ENVIRONMENT	Set to 'production' on Railway
Sentry release	RELEASE_VERSION	Optional: set on each deploy for release tagging
UptimeRobot	(no env var — sign up)	uptimerobot.com → Add Monitor → ping https://yourdomain.com/api/health
Headers debug	CSP_REPORT_ONLY=true	Set temporarily in Railway if a 3rd-party widget breaks

This document is intentionally organised by **priority of harm if missed**, not by ease of build. If you only ship Tier 1 + Tier 2 + Tier 6, you've covered the launches that matter most.